CODERSINPUNE 2026 · ROUND 2: THE BUILD

ReV — Handover Engine

Evidence-Backed Knowledge Transfer & Developer Offboarding Platform

Team:	**ReV Team**
Live Link:	https://rev.fev06.xyz
Source Code:	github.com/FeV-06/ReV (Public)
AI Architecture:	**Exact Substring Citations & Grounded Reasoning**

1. Executive Summary & Core Mission

When an experienced engineer leaves a software team, they take weeks of undocumented architectural decisions, recovery procedures, and edge-case warnings with them. Traditional handovers fail because static markdown docs become stale within days, and free-form chats suffer from severe hallucinations.

ReV (Relaying to Everyone Virtually) solves this crisis by transforming fragmented commit histories, Tree-Sitter AST code symbols, and Slack discussions into an **evidence-linked, human-verified developer handover**. ReV enforces mathematical citation integrity: every claim must cite an exact substring from an authorized source artifact.

2. System Architecture: The Three-Tier Local-First Model

ReV is engineered as a hybrid local-first system that decouples native CPU-bound code analysis from client presentation and persistent team synchronization:

1. Headless Daemon :3078

Node 22+ background service bound to loopback. Owns persistent PGlite relational storage, orchestrates Git watchers, and executes native Rust core tasks.

2. Web Dashboard :3077

Next.js 16 / React 19 UI following Google's **Material 3 Expressive** tokens. Features a 4-stage screen-advancing Studio and Successor Reader.

3. VS Code Extension IDE NATIVE

Dual-bundle extension (Node host + Browser IIFE) embedding the shared React Team Sync panel and workspace-scoped telemetry inside the developer's editor.

Native Rust Core Engine (crates/rev-core) HIGH PERFORMANCE

Compiled native binary that executes Tree-Sitter AST symbol parsing (TypeScript, Rust, and Python fallback), git blame ownership and churn heuristics (`analyze_local_risk`), and verifies verbatim citation quotes before generating candidates.

3. Completed Round 2 Feature Deliverables

- ✔ **Dedicated Dual-Mode Interface:** Zero-friction toggle between ⚡ *Instant Demo* (0s latency, interactive spotlight tour) and 🔬 *Live Engine* (real public GitHub repos, Tree-sitter AST, live AI reasoning).
- ✔ **4-Stage Material 3 Creation Wizard:** Sequential screen-advancing studio (Workspace & Contributor ➔ Scope Confirmation ➔ Synthesis & Interview ➔ Export & Sovereignty).
- ✔ **Tree-Sitter AST Code Symbol Mapping:** Extracts changed functions, classes, and types from diffs, linking AST symbols directly to Git commit authors.
- ✔ **Grounded Departure Interview Engine:** Core generates standalone gap questions by cross-referencing commit history with repository documentation.
- ✔ **Interactive Successor Q&A Assistant:** Answers successor questions by citing exact verbatim excerpts from codebase artifacts and interview answers.
- ✔ **Priority Day-1 Onboarding Checklist:** Successor portal task tracker for verified codebase setup, credentials, and deployment runbooks with local state.
- ✔ **Local Data Sovereignty Controls:** Built on local PGlite relational storage by default for absolute data privacy, with staging controls and UI toggles for upcoming Supabase team sync.
- ✔ **Multi-Source Scoping & Granular Invalidation:** Real-time selection of commits and Slack messages; scope edits atomically invalidate prior approvals.
- ✔ **One-Click Cited Markdown Export:** Generates publication-ready `HANDOVER.md` complete with verifiable source citations and next steps.
- ✔ **1-Click Benchmark Presets:** Pre-configured evaluation shortcuts for real open-source repositories (`KingSahil/Blinky`, `FeV-06/ReV`).

4. What Changed Since Round 1 (And Why)

Rather than treating Round 1 as an unchangeable specification, we evolved ReV based on realistic operational challenges encountered during early testing:

Pillar / Area	Round 1 Proposal	Shipped in Round 2 (The Build)
System Architecture	Single monolithic web form running purely in-browser against local files.	**Decoupled Three-Tier Architecture:** Headless Desktop Daemon (:3078), Next.js M3 Web Studio (:3077), and native VS Code Extension.
Database & Sync	Local-only single-session storage; no sharing between team members.	**PGlite Storage + Staged Supabase Sync:** Full local PGlite relational engine active today; multi-tenant Supabase schemas and RLS policies are implemented and verified in isolated test suites, staged for Round 3 rollout.
Handover Generation	Static LLM text reconstruction without active questioning or follow-ups.	**Departure Interviews & Cited Q&A:** Proactive gap generator asking departing engineers tough architectural questions, plus successor assistant.
User Experience	Stacked single-page forms with developer persona buttons cluttering views.	**Material 3 Expressive System:** Clean role-tailored portals (Admin Studio vs. Successor Onboarding Reader) with animated slide transitions.
Code Intelligence	Basic commit message and diff grep regex matching.	**Tree-Sitter AST Analysis:** Multi-language AST parsing extracting exact functions, structs, and methods directly impacted by commit churn.

5. Transparent Operational Disclosures ("What Does NOT Work Yet")

Transparency Notice: ReV is an evidence-backed product. Below is the honest, unvarnished status of work-in-progress boundaries:

- **Supabase Team Sync is In-Progress (WIP):** The multi-tenant architecture, PostgreSQL RLS migrations, and idempotent RPCs (`rev_create_team`, `rev_invite_member`, `rev_publish_document`) are implemented and pass 38 isolated security and unit tests. However, live multi-tenant route authorization across all Next.js API endpoints is actively under development. To protect data sovereignty and prevent unauthorized cross-session leakage, live remote network synchronization is deliberately kept fail-closed, and legacy bulk sync is disabled (HTTP 403). Full frictionless team sync is our core Round 3 commitment.
- **Live Slack OAuth Configuration:** Connecting a live Slack workspace requires pre-configured OAuth bot client credentials. For zero-config evaluation, ReV provides full synthetic ZIP archive ingestion.
- **Unauthenticated GitHub API Bounds:** Without a configured GitHub personal access token (`REV_GITHUB_TOKEN`), public repo imports are constrained by GitHub's 60 requests/hour IP rate limit.
- **Deep Reasoning Latency:** Synthesizing comprehensive handovers with deep reasoning models takes 45–90 seconds over large symbol trees. The UI provides live multi-phase progress indicators with a live timer, and instant zero-latency evaluation is accessible via the synthetic showcase demo.

6. Binding Commitments for Round 3 (The Finish)

We commit to delivering the following three substantial, technically rigorous improvements for Round 3:

Commitment 1: Laya Classifier — AI PIPELINE

Commit & AST Semantic Triage: Integrate a specialized classifier

Commitment 2: Team Sync — MULTI-TENANT

Frictionless Supabase Team Sync: Complete the end-to-end

Commitment 3: Risk Heatmap — RISK ENGINE

Bus-Factor & Churn Heatmap: Deliver an interactive visual risk

pipeline to evaluate whether an AST symbol or commit represents an intentional architectural decision versus routine cosmetic churn, pruning noise before LLM synthesis.

authenticated team mode with verified member sessions, private-by-default owned runs, and frictionless one-click lead sharing across the web dashboard and VS Code extension.

heatmap identifying repository modules with critical knowledge concentration (combining Git blame ownership, commit churn, and test coverage gaps via native Rust heuristics).

Judges Verification Guide: Open **https://rev.fev06.xyz** in any modern browser. You can click *"Try sample project"* to run through the complete 4-stage wizard, inspect AST symbol citations, and run the departure interview with zero setup, or provide any public GitHub repo to experience live model inference.